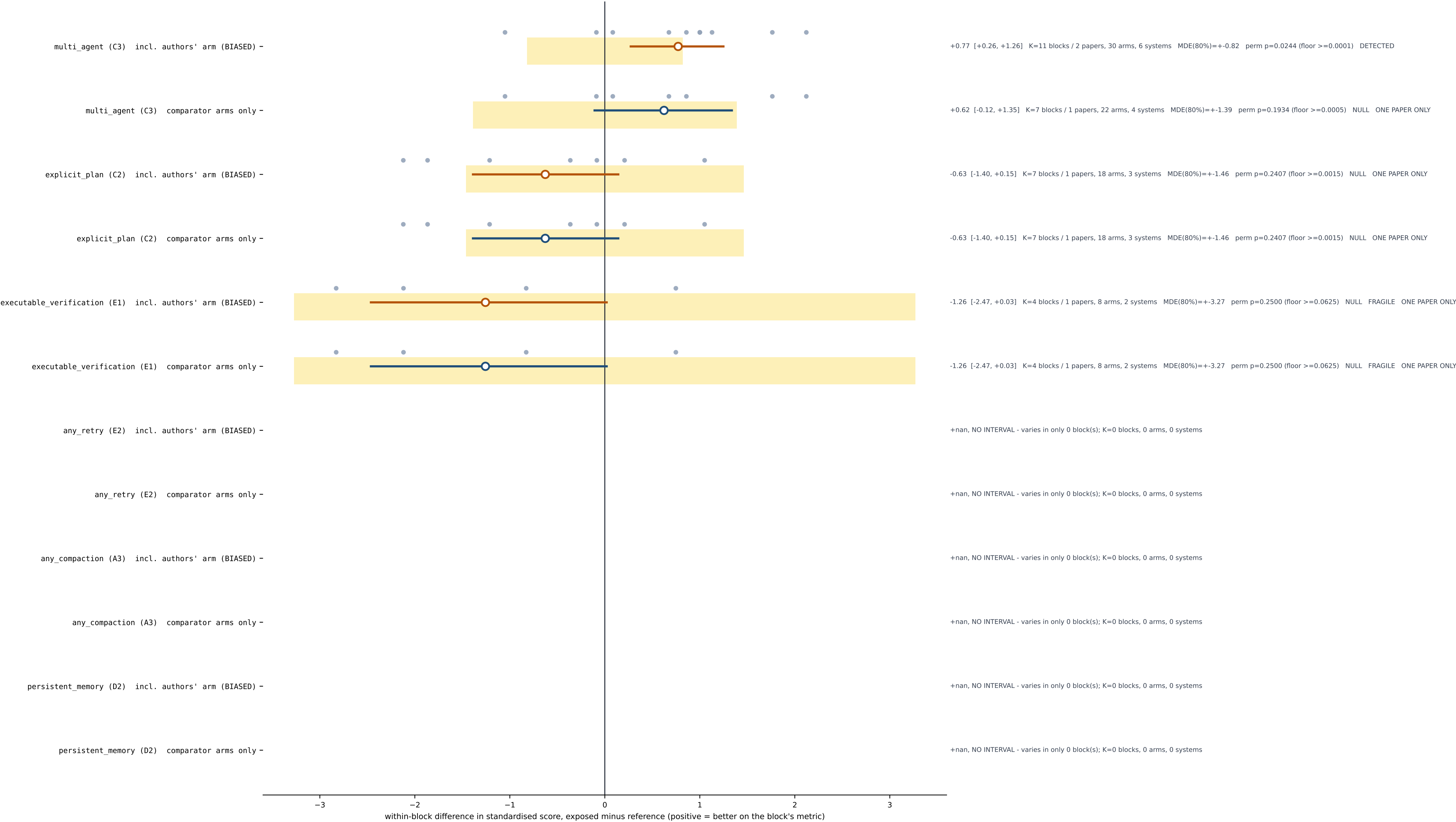


Pre-specified harness-design contrasts inside one paper's own comparison table
Blue: the headline, comparator arms only, the authors' own arm excluded. Amber: the same contrast with the authors' arm included - OPTIMISTICALLY BIASED, shown for comparison only.
Open circle: mean of block-level differences. Bar: percentile bootstrap CI resampling BLOCKS. Shaded band: +- the minimum detectable effect at 80% power (amber band = only a large effect was detectable).
Small grey dots: the individual block-level differences the estimate is made of.



All contrasts were fixed as a set before estimation and all are reported, including the null ones and the ones the data cannot support; nothing was dropped for its result. A block contributes only if the contrast varies inside it, which is what makes each estimate a within-block one, and the paper is the blocking factor. A CI covering zero means no detectable association at this block count - read it with the minimum detectable effect on the same line, which is what could have been found. OWN-ARM BIAS, the confound that dominates this design: in a paper's own baseline table the authors' system is the one they tuned, while the comparators are usually run at default settings by people who did not build them, and the table exists to show the authors' system winning. 'The paper's own system wins' is therefore close to tautological. The headline estimates here EXCLUDE the authors' own arm and compare comparator arms with each other; the version including that arm is reported separately and is OPTIMISTICALLY BIASED. Scaffold drift (docs/benchmark_caveats.md sec. 5) is reduced but not removed: the block holds one team, one protocol, one benchmark, one split, one metric and one model fixed, so the cross-paper protocol difference is gone, but the rival was run by the reporting paper at some commit while HARNESS-DB codes it at a pinned commit, which may be a different one. That is measurement error on the design variable; it attenuates towards zero, so every effect is a LOWER BOUND on magnitude. Reporting is also self-selected (sec. 3): which rivals appear in a table is the authors' choice. Associational only; no causal reading is supported.

Power, computed from the observed between-block spread and from the design itself.
- multi_agent: with 11 blocks from 2 papers, 30 arms and 6 distinct coded systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 0.82 within-block sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. Only 2 papers contribute, so the between-paper replication the design relies on is minimal. The design admits 32,768 distinct within-block label assignments, so the smallest attainable permutation p-value is 0.0001 (observed 0.0244).
- multi_agent: with 7 blocks from 1 papers, 22 arms and 4 distinct coded systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 1.39 within-block sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. NOT INDEPENDENTLY REPLICATED: all 7 blocks come from ONE paper, so they share the authors, the implementation of every comparator and usually one table. The paper is the blocking factor and there is only one of it; the interval describes that table, not the field, and nothing here can be called a detected association. The design admits 2,048 distinct within-block label assignments, so the smallest attainable permutation p-value is 0.0005 (observed 0.1934).
- explicit_plan: with 7 blocks from 1 papers, 18 arms and 3 distinct coded systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 1.46 within-block sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. NOT INDEPENDENTLY REPLICATED: all 7 blocks come from ONE paper, so they share the authors, the implementation of every comparator and usually one table. The paper is the blocking factor and there is only one of it; the interval describes that table, not the field, and nothing here can be called a detected association. The design admits 648 distinct within-block label assignments, so the smallest attainable permutation p-value is 0.0015 (observed 0.2407).
- explicit_plan: with 7 blocks from 1 papers, 18 arms and 2 distinct coded systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 1.46 within-block sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. NOT INDEPENDENTLY REPLICATED: all 7 blocks come from ONE paper, so they share the authors, the implementation of every comparator and usually one table. The paper is the blocking factor and there is only one of it; the interval describes that table, not the field, and nothing here can be called a detected association. The design admits 648 distinct within-block label assignments, so the smallest attainable permutation p-value is 0.0015 (observed 0.2407).
- executable_verification: with 4 blocks from 1 papers, 8 arms and 2 distinct coded systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 3.27 within-block sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. FRAGILE: fewer than 5 blocks, so the between-block spread that both the interval and this number rest on is itself barely estimated; treat the interval as descriptive. NOT INDEPENDENTLY REPLICATED: all 4 blocks come from ONE paper, so they share the authors, the implementation of every comparator and usually one table. The paper is the blocking factor and there is only one of it; the interval describes that table, not the field, and nothing here can be called a detected association. The design admits 16 distinct within-block label assignments, so the smallest attainable permutation p-value is 0.0625; no conventional significance level is reachable from this contrast whatever the point estimate.
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- any_retry: NOT ESTIMABLE - the contrast varies inside only 0 block(s), so no interval and no minimum detectable effect are defined.
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- any_compaction: NOT ESTIMABLE - the contrast varies inside only 0 block(s), so no interval and no minimum detectable effect are defined.
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- persistent_memory: NOT ESTIMABLE - the contrast varies inside only 0 block(s), so no interval and no minimum detectable effect are defined.
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